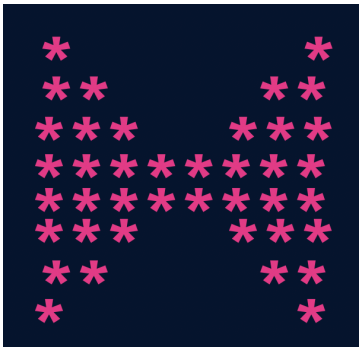
# Java - Introduction to Programming

# Lecture 6

## Patterns - Part 2

1. 

```java
import java.util.*;

public class Solutions {
   public static void main(String args[]) {
       int n = 4;

       //upper part
       for(int i=1; i<=n; i++) {
           for(int j=1; j<=i; j++) {
               System.out.print("*");
           }

           int spaces = 2 * (n-i);
           for(int j=1; j<=spaces; j++) {
               System.out.print(" ");
           }

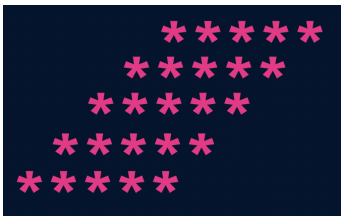
           for(int j=1; j<=i; j++) {
               System.out.print("*");
           }
           System.out.println();
       }
```

```
        //lower part
        for(int i=n; i>=1; i--) {
            for(int j=1; j<=i; j++) {
                System.out.print("*");
            }

            int spaces = 2 * (n-i);
            for(int j=1; j<=spaces; j++) {
                System.out.print(" ");
            }

            for(int j=1; j<=i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

2.




```
import java.util.*;

public class Solutions {
    public static void main(String args[]) {
        int n = 5;
```

```
        for(int i=1; i<=n; i++) {
            //spaces
            for(int j=1; j<=n-i; j++) {
                System.out.print(" ");
            }

            //stars
            for(int j=1; j<=n; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

3.



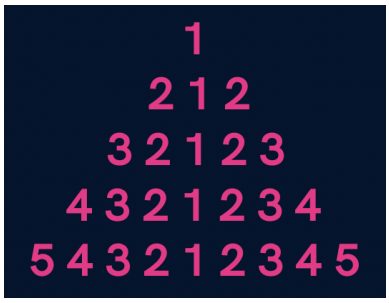

```java
import java.util.*;

public class Solutions {
   public static void main(String args[]) {
       int n = 5;

       for(int i=1; i<=n; i++) {
           //spaces
           for(int j=1; j<=n-i; j++) {
               System.out.print(" ");
           }

           //numbers
           for(int j=1; j<=i; j++) {
               System.out.print(i+" ");
           }
           System.out.println();
       }
   }
}
```

4.



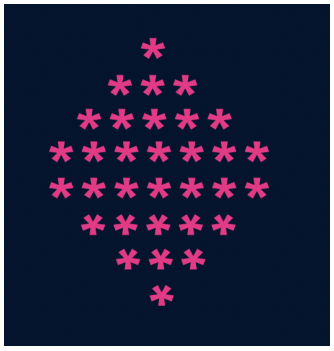

```java
import java.util.*;

public class Solutions {
   public static void main(String args[]) {
       int n = 5;
       for(int i=1; i<=n; i++) {
           //spaces
           for(int j=1; j<=n-i; j++) {
               System.out.print(" ");
           }

           //first part
           for(int j=i; j>=1; j--) {
               System.out.print(j);
           }

           //second part
           for(int j=2; j<=i; j++) {
               System.out.print(j);
           }
           System.out.println();
       }
   }
}
```

5.



```java
import java.util.*;

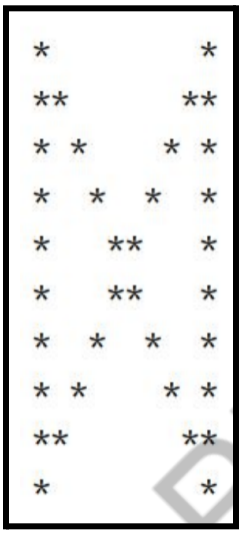
public class Solutions {
   public static void main(String args[]) {
       int n = 5;

       //upper part
       for(int i=1; i<=n; i++) {
           //spaces
           for(int j=1; j<=n-i; j++) {
               System.out.print(" ");
           }
           for(int j=1; j<=2*i-1; j++) {
               System.out.print("*");
           }
           System.out.println();
       }

       //lower part
       for(int i=n; i>=1; i--) {
           //spaces
           for(int j=1; j<=n-i; j++) {
               System.out.print(" ");
           }
           for(int j=1; j<=2*i-1; j++) {
               System.out.print("*");
           }
           System.out.println();
       }
   }
}
```

## Homework Problems

1. **Print a hollow Butterfly.**



2. **Print a hollow Rhombus.**

```
  *****
 *     *
*     *
*     *
*****
```

3. **Print Pascal's Triangle.**

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

4. **Print half Pyramid.**

```
    1
```

1 2

1 2 3

1 2 3 4

1 2 3 4 5

5. **Print Inverted Half Pyramid.**

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4